M1 - AAGB - BIM

Algorithms for the analysis of genome rearrangements

Alessandra Carbone
Sorbonne Université

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Turnip and cabbage are different in appearence and taste, but they share a recent ancestor

Sequence comparison of genes does not provide interesting evolutionary information

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Turnip vs Cabbage : genes of mitochondrial DNA are almost identical

- In 1980s Jeffrey Palmer studied organelles evolution in plants by comparing mitochondrial genomes of turnip and cabbage.
- 99% of similarity among genes
- These sequences were different in the order of their genes. This came as a surprise!
- This study started a systematic analysis of genome rearrangements in molecular evolution.

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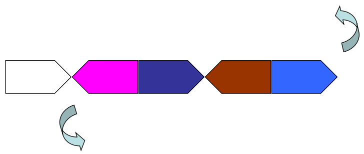
Turnip vs cabbage: different order of genes in mtDNA

Comparison of gene order:



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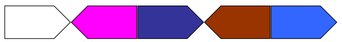

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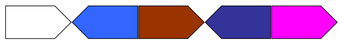

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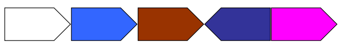

Turnip vs cabbage: different order of genes in mtDNA

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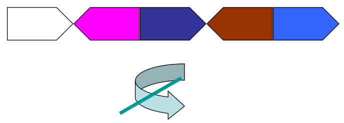
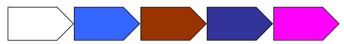


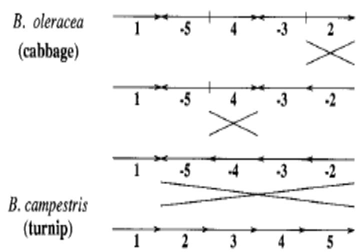
Cabbage



Turnip

Evolution is present in the divergence of gene order

Transformation of a cabbage into a turnip



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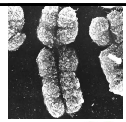
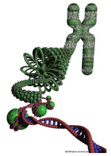
Operations on chromosomes





On a single chromosome

Deletion : a part is lost abc → ac, where a,b,c are sub-sequences

Insertion : a part is added ac → abc

Duplication : a part is duplicated abc → abbc, tandem
abcd → abcbd or abcd → acbcd

Inversion/reversal : a part is inverted, head-tail $abc_1c_2c_3c_4de \rightarrow ab\text{-}c_4\text{-}c_3\text{-}c_2\text{-}c_1de$

Transposition : two adjacent parts exchange their position : abcd → acbd

This operation is probably the most rare because it demands at least 3 breakpoints on the chromosome

Transversal : two parts exchange their position and one is inversed

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We do not know how these operations are physically implemented in the cell.

If two regions on the chromosome are sufficiently similar, then we think that they can hybridize (similarly to two DNA sequences forming a double helice).

When they hybridize there is a formation of a loop. This loop can be eliminated (deletion), or its direction can change (inversion/reversal).

An insertion can occur when a new DNA part enters in the cell (during a viral infection or an horizontal transfert) and it recombines with the DNA of the cell.

A duplication can happen during mitosis by mistake.

Transpositions are rare, and we have no idea about their formation.

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Example of Rearrangements

Reversal/Inversion

1 2 3 4 5 6 → 1 2 -5 -4 -3 6

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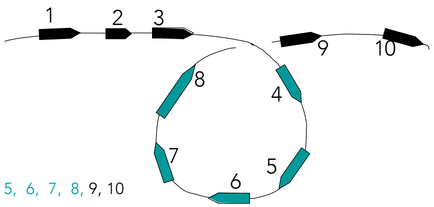
Reversal



1, 2, 3, 4, 5, 6, 7, 8, 9, 10

- Blocs represent conserved genes

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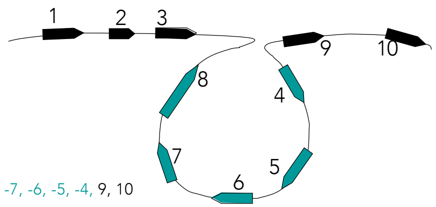
Reversal



1, 2, 3, -8, -7, -6, -5, -4, 9, 10

- Blocs represent conserved genes.
- During evolution or in clinical contexts (e.g. cancer), blocks 1,…,10 can be erroneously read as 1, 2, 3, -8, -7, -6, -5, -4, 9, 10.

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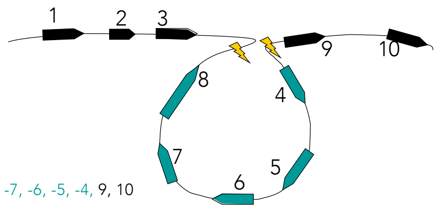
Reversal and breakpoints



1, 2, 3, -8, -7, -6, -5, -4, 9, 10

Reversal introduced two *breakpoints*
(inducing a change in the order).

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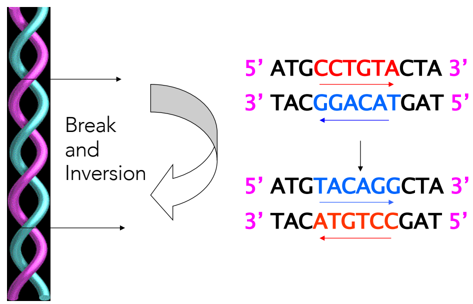
Reversal: operation over two strands



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Operations on two chromosomes

Translocation : two chromosomes exchange their tails.

Attention: not all translocations are possible. A chromosome contains a part called centromere that is crucial during the process of cell division. The centromere is usually localized at the center of a chromosome, and if after translocation its position is lost in one of the chromosomes, the cell can die.

Fusion : two chromosomes mix in one single chromosome

Fission : a chromosome splits into two chromosomes.

It is not known what happens with centromeres in the last two cases.

Examples of Rearrangements

Translocation

1 2 3 → 1 2 6
4 5 6 → 4 5 3

Fusion

1 2 3 4
5 6 ⇄ 1 2 3 4 5 6

Fission

Genome rearrangements and evolution

There exist several factors that render the study of rearrangements important for evolution.

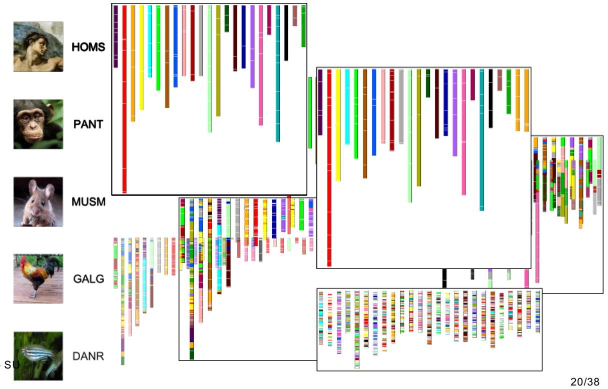
Rearrangement operations are much more rare than pointwise mutations. This implies that we can try to trace genome rearrangements during their evolution, much further back than for regular mutations.

For instance, by compairing man and mouse with about 75 millions years of evolutionary distance among the two, we observe only about 200 rearrangement operations.

There is a very small possibility of « reversed mutations » to happen (that is a backward mutation occurring twice on a genome, twice in the same spot). Therefore, we have a weak ambiguity in the interpretation of these mutations.

Since rearrangements are a kind of macroscopic mutations, we can localize them in rather large portions of genomes and analyze them at a larger scale. This allows to render more accessible/clear our understanding of the evolutionary process.

Evolution of species and chromosomal rearrangements



Genome rearrangements

Unknown ancestor
~ 75 million years ago
(evolutionary distance between man and mouse)

Mouse

Human

But only 245 operations of chromosomal rearrangements

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Genome rearrangements

Unknown ancestor
~ 75 million years ago
(evolutionary distance between man and mouse)

Mouse

Human

But only 245 operations of chromosomal rearrangements

- What are similar blocks and how to find them?

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Genome rearrangements

Unknown ancestor
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Mouse

Human

But only 245 operations of chromosomal rearrangements

- What are similar blocks and how to find them?
- What is the architecture of the ancestral genome?

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Genome rearrangements

Unknown ancestor
~ 75 million years ago
(evolutionary distance between man and mouse)

Mouse

Human

But only 245 operations of chromosomal rearrangements

- What are similar blocks and how to find them?
- What is the architecture of the ancestral genome?
- What is the evolutionary scenario that justifies the transformation of a genome into another ?

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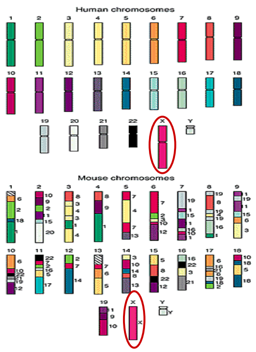
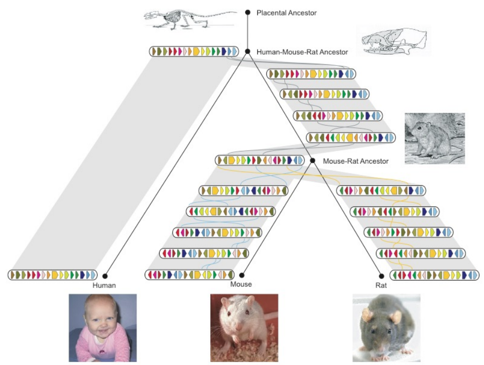
Comparison of the genome architectures for man and mouse

- Humman and mouse have a similar genome, but their genes are ordered differently
- ~245 rearrangements
  - Reversals
  - Fusions
  - Fissions
  - Translocations

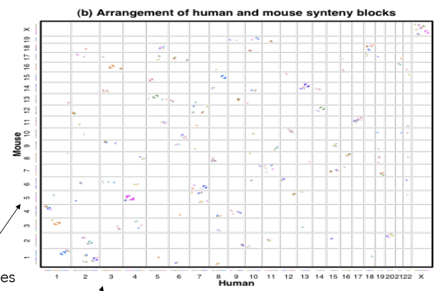


Chromosome X



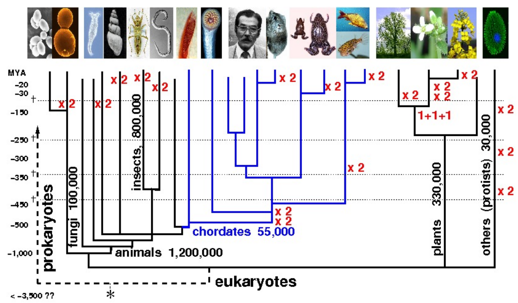
Rat Consortium, *Nature*, 2004

Comparison of the genome architectures for man and mouse through dotplots

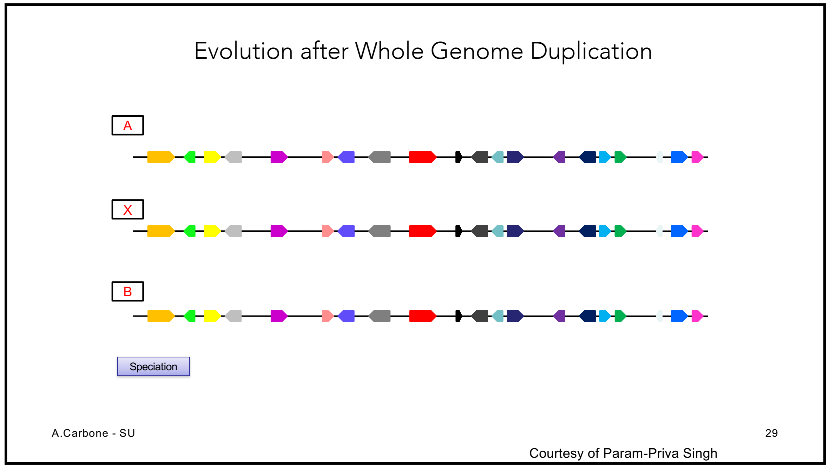


chromosomes

Genome Duplications on the Tree of Life



WGD gave rise to increased genomic & morphological complexity in vertebrates.


Evolution after Whole Genome Duplication
A
X
B
Speciation
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Courtesy of Param-Priva Singh

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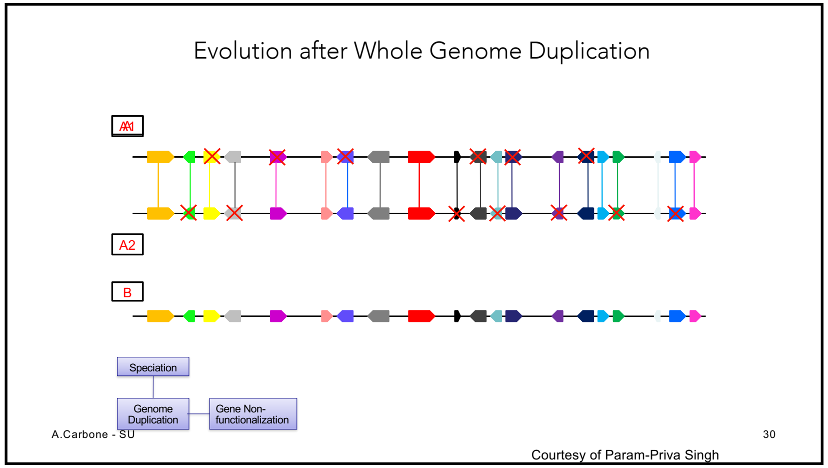
Evolution after Whole Genome Duplication
A1
A2
B
Speciation
Genome Duplication
Gene Non-functionalization
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Courtesy of Param-Priva Singh

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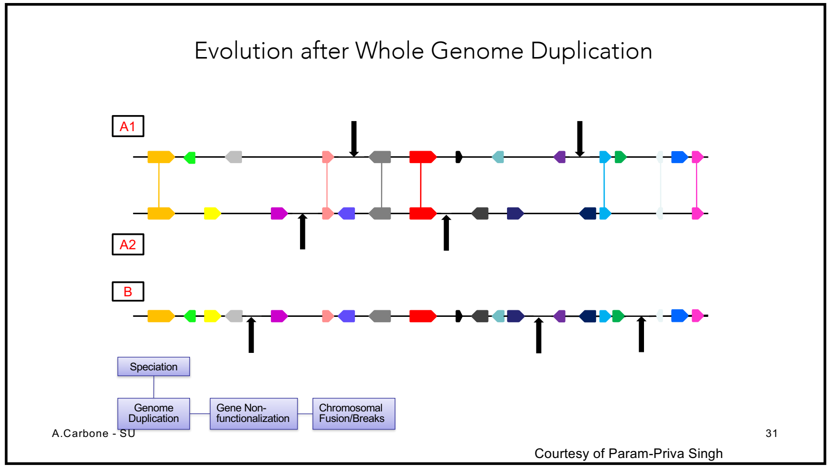
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Chromosomal Fusion/Breaks
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Courtesy of Param-Priva Singh

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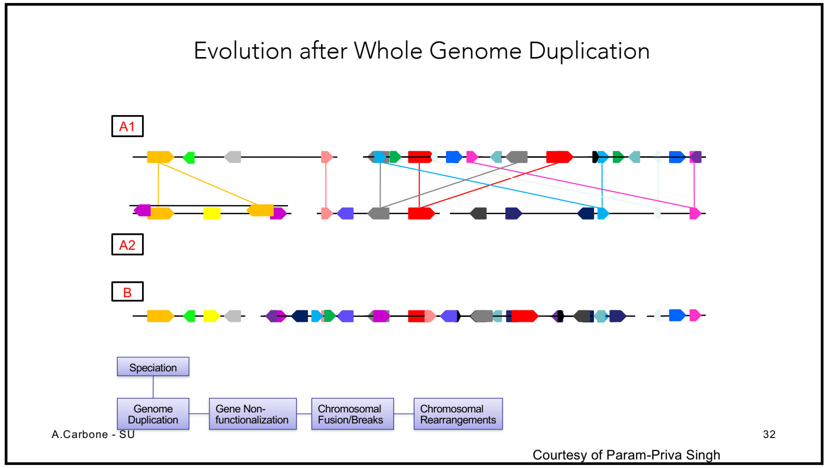
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Chromosomal Rearrangements
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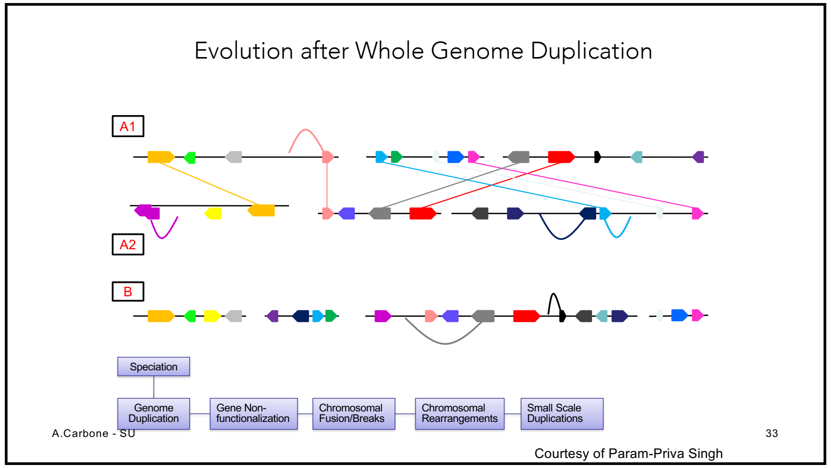
Evolution after Whole Genome Duplication
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Small Scale Duplications
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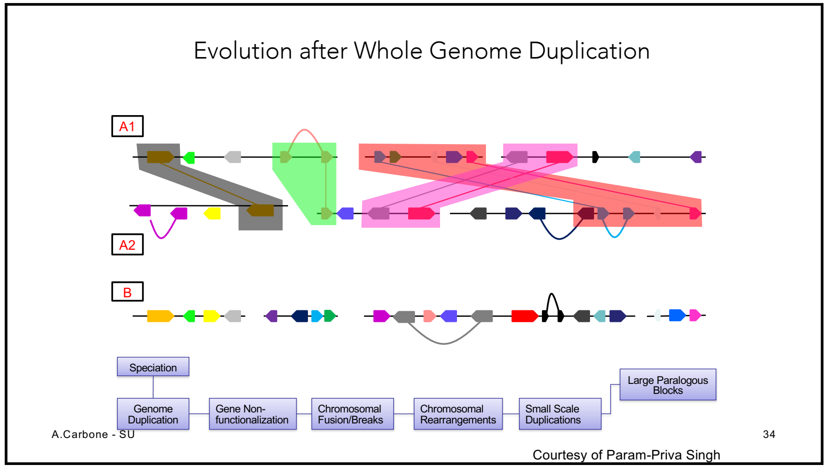
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Chromosomal Rearrangements
Small Scale Duplications
Large Paralogous Blocks
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Courtesy of Param-Priva Singh

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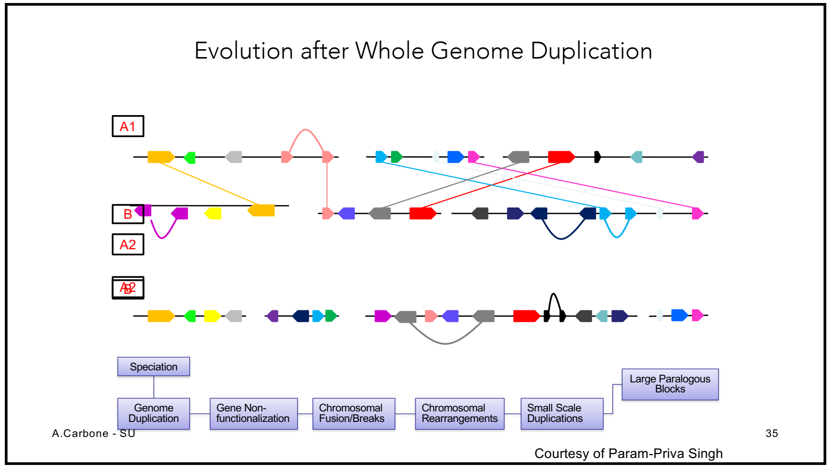
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Large Paralogous Blocks
A.Carbone - SU
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Courtesy of Param-Priva Singh

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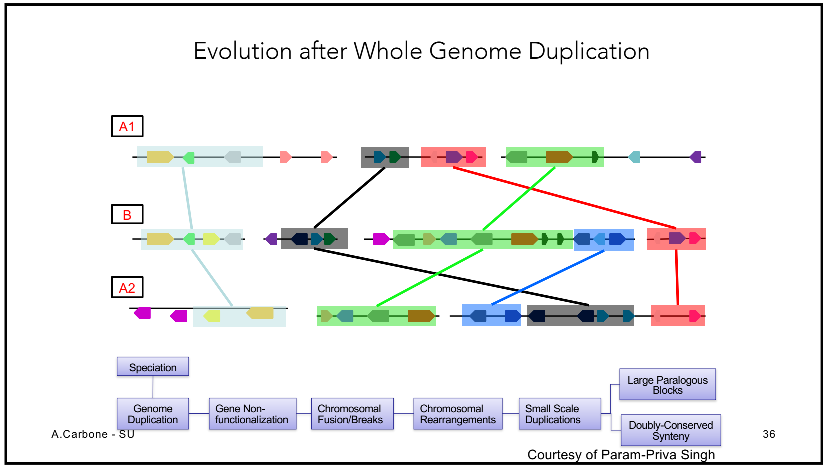
Evolution after Whole Genome Duplication
A1
B
A2
Speciation
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Gene Non-functionalization
Chromosomal Fusion/Breaks
Chromosomal Rearrangements
Small Scale Duplications
Large Paralogous Blocks
Doubly-Conserved Synteny
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Courtesy of Param-Priva Singh

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Reversal : an example

$$\pi = 1\ 2\ \underline{3\ 4\ 5}\ 6\ 7\ 8$$

$\rho(3,5)$ positions

$$1\ 2\ 5\ 4\ 3\ 6\ 7\ 8$$

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Reversal : an example

$$\pi = 1\ 2\ \underline{3\ 4\ 5}\ 6\ 7\ 8$$

$\rho(3,5)$ positions

$$1\ 2\ 5\ 4\ \underline{3\ 6}\ 7\ 8$$

$\rho(5,6)$

$$1\ 2\ 5\ 4\ 6\ 3\ 7\ 8$$

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Reversal and gene order

In general, gene order is represented by a permutation $\boldsymbol{\pi}$:

$$\pi\ = \pi_1 \text{------} \underline{\pi_{i-1}\, \pi_i\, \pi_{i+1} \text{------} \pi_{j-1}\, \pi_j}\, \pi_{j+1} \text{-----} \pi_n$$

$\rho(i,j)$

$$\pi_1 \text{------} \pi_{i-1}\, \pi_j\, \pi_{j-1} \text{------} \pi_{i+1}\, \pi_i\, \pi_{j+1} \text{-----} \pi_n$$

The reversal $\rho(\, i, j\,)$ inverts the elements from i to j in π

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Unsigned permutations

Study of inversions on unsigned permutations.

Hypotheses:

1. We shall consider a single chromosome and we shall work on two hypotheses :

- It is possible to identify genes without ambiguity on chromosomes and to identify their homologues in other species.
- All genes are different

2. Genes in a sequence will be represented by elements of the permutation and the direction of each gene is forgotten.

3. Gene order (or the order of their homologues), possibly different in different species, is a permutation of these genes (or of their homologues):
a permutation $\pi = (\pi_1, \ldots \pi_n)$ of unsigned integers will represent orders of different genes

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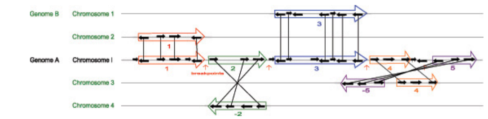
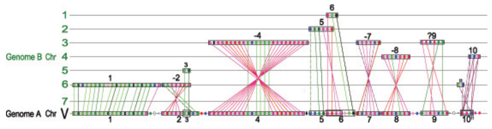
With multiple chromosomes



Ideal case (with multiple chromosomes)



Real case in yeast species

Definition : a transformation of reversals on a sequence is the operation that takes a subsequence and inverts the order of the elements. For example 1$\underline{2345}$ → 14325.

Definition : a reversal distance between two sequences is the minimum number of reversals that are necessary to transform one into the other.

Example:

Reversal distance = 3

π_1 = (1 $\underline{2\ 3\ 4}$ 5 6)
π_2 = (1 4 $\underline{3\ 2\ 5\ 6}$)
π_3 = ($\underline{1\ 4\ 6}$ 5 2 3)
π_4 = (6 4 1 5 2 3)

The underlined subwords correspond to the position of the reversal

Problem (ordering by reversal)
Input: a permutation π on n elements
Output: the reversal distance between π and the identity permutation **(id = (1 2 ... n))**.

Problem (equivalent to the previous one?) :
Given two permutations, find the shortest series of reversals that transforms one into the other.
Entrée: Permutations π and σ
Sortie: A series of reversals $r_1,\ldots,r_t$ transforming π in σ, such that t (reversal distance between π and σ) is minimal.

For example:

$$\pi = (7\,3\,2\,4\,5\,1\,6) \longrightarrow \sigma = (2\,3\,5\,1\,4\,6\,7)$$

$$\Downarrow$$

We take $\sigma = (2\ 3\ 5\ 1\ 4\ 6\ 7)$, we rename it

$(1'\ 2'\ 3'\ 4'\ 5'\ 6'\ 7')$

and we rewrite the problem as follows:

$$\pi = (7'\ 2'\ 1'\ 5'\ 3'\ 4'\ 6') \longrightarrow \sigma = (1'\ 2'\ 3'\ 4'\ 5'\ 6'\ 7')$$

Ordering by reversals: 4 steps

Step 0: π	2	4	3	5	8	7	6	1
Step 1:	2	3	4	5	8	7	6	1
Step 2:	5	4	3	2	8	7	6	1
Step 3:	5	4	3	2	1	6	7	8
Step 4: γ	1	2	3	4	5	6	7	8

What is the reversal distance for this permutation ?
Can we order it with 3 steps ?

Several solutions have been proposed:

Algorithm of 2-approximation	(Kececioglu and Sankoff, 1995)
Algorithm of 1.75-approximation	(Bafna and Pevzner, 1996)
Algorithm of 1.5-approximation	(Christie, 1998)
Algorithm of 1.375-approximation	(Hannenhalli, 1998)
Proof of NP-completness	(Caprara, 1999)

We start with solutions that use greedy algorithms

Pancake Flipping Problem

- The boss prepares a tower of pancakes which are unordered in size
- The servant wishes to order them (in such a way that the smallest is on the top and the largest on the bottom)
- He will do it by reversing several pancakes from the top, and by repeating the idea sufficiently many times.

Christos Papadimitrou and Bill Gates reversing pancakes

Problem (pancakes flipping) : Given a tower of n pancakes, what is the minimal number of reversals to obtain a perfect tower?
Input: Permutation π
Output: a series of prefix reversals r_1, … n transforming π into the identity permutation such that t is minimal.

Greedy approach : we need at most 2 prefix reversals to place a pancake at its correct position. Hence we need at most 2n – 2 steps in total.

Bill Gates and Christos Papadimitriou showed in the 70s that any permutation can be ordered in at most 5/3 (n + 1) prefix reversals. On the other hand, the problem of pancakes flipping remains unsolved (that is, the search of a solution with the smallest number of reversals).

Order by reversals : a greedy algorithm

- If we order π = 1 2 3 6 4 5, the first three elements are already on the right order, hence we need not to move them nor to separate them.
- The length of an existing prefix of π is denoted *prefix*(π)
  - *prefix*(π) = 3
- This observation is at the base of the greedy algorithm: increase *prefix*(π) at each step.

Example: π will be ordered as follows

1 2 3 <u>6 4</u> 5

1 2 3 4 <u>6 5</u>

1 2 3 4 5 6

The number of steps to order a permutation of length n is at most *(n – 1)*

Greedy Algorithm: pseudocode

```
SimpleReversalSort(π)
for i ← 1 to n – 1
  j ← position of element i in π (i.e., πj = i)
  if j ≠ i
    π ← π · ρ(i, j)
    output π
  if π is the identity permutation
   return
```


Analysis of SimpleReversalSort

SimpleReversalSort does not guarantee the smallest number of reversals and takes 5 steps on π = 6 1 2 3 4 5 :

- Step 1: 1 6 2 3 4 5
- Step 2: 1 2 6 3 4 5
- Step 3: 1 2 3 6 4 5
- Step 4: 1 2 3 4 6 5
- Step 5: 1 2 3 4 5 6

The permutation of the example (π = 6 1 2 3 4 5) can be ordered in 2 steps

Step 1: 5 4 3 2 1 6
Step 2: 1 2 3 4 5 6

Then, SimpleReversalSort(π) is not optimal

The permutation of the example (π = 6 1 2 3 4 5) can be ordered in 2 steps

Step 1: 5 4 3 2 1 6
Step 2: 1 2 3 4 5 6

Then, SimpleReversalSort(π) is not optimal

Optimal algorithms are unknown; algorithms approximating the optimal solution are used.

These algorithms find more often approximating solutions than optimal solutions. The approximation ratio of an algorithm A on input π is:

$$A(\pi) / OPT(\pi)$$

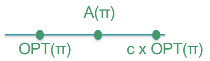
where
A(π) – solution obtained with algorithm A
OPT(π) – optimal solution

The approximation (performance guarantee) of algorithm A:
max/min of the approximation ratio on all inputs of size *n*

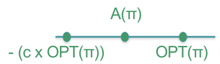
$\max_{|\pi| = n} A(\pi) / OPT(\pi)$ for minimizing algorithms A
$\min_{|\pi| = n} A(\pi) / OPT(\pi)$ for maximizing algorithms A

For each instance π of the problem P, we search for a solution of value A(π) such that:

OPT(π) ≤ A(π) ≤ c x OPT(π) if P is a minimization problem



-(c x OPT(π)) ≤ A(π) ≤ OPT(π) if P is a maximization problem



We call c the approximation ratio when it satisfies inequalities for all the entries π of size n

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If you are interested

To catch up with the notion of approximation algoritms, see for example:

https://www.youtube.com/watch?v=g-NLgxhCW-A

Course by Frédéric Vivien : Algorithmes d'approximation - Part 1 delivered at the CIRM in Marseille.

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Adjacencies and breakpoints

$$\pi = \pi_1\pi_2\pi_3\ldots\pi_{n-1}\pi_n$$

A pair of elements π_i and π_{i+1} are adjacent if

$$\pi_{i+1} = \pi_i \pm 1$$

For example:

π = 1 9 <u>3 4</u> <u>7 8</u> 2 <u>6 5</u>

(3, 4), (7, 8) and (6,5) are adjacent pairs.

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Adjacencies and breakpoints

Definition : a breakpoint is any point in the sequence where two adjacent numbers are not consecutive ($|\pi_i - \pi_{i+1}| \neq 1$).

Example : in the sequence 123|654 there is a breakpoint between 3 and 6.

Example:

π = 1 | 9 | 3 4 | 7 8 | 2 | 6 5

Pairs (1,9), (9,3), (4,7), (8,2) and (2,6) form the breakpoints of the permutation π

$b(\pi)$ - # of breakpoints in the permutation π

"b" symbolizes the "breakpoints"

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Adjacencies and breakpoints

- An adjacency is a pair of adjacent consecutive elements
- A breakpoint is a pair of adjacent non consecutive elements

Example:

$\pi = 5\ 6\ 2\ 1\ 3\ 4$

adjacencies

5 6 2 1 3 4

breakpoints

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Attention: extension of permutations

We insert two elements $\pi_0 = 0$ and $\pi_{n+1} = n+1$ to the ends of π

Example:

π = 1 |9| 3 4| 7 8|2 |6 5

extend with 0 and 10

π = 0 1|9| 3 4| 7 8|2|6 5|10

Note: a new breakpoint has been introduced

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Lemma (reversal distance and breakpoints) : each reversal eliminates at most 2 breakpoints.

π = 2 3 1 4 6 5

0 |2 3|1 |4| 6 5| 7	$b(\pi) = 5$
0 1 |3 2 |4| 6 5| 7	$b(\pi) = 4$
0 1 2 3 4| 6 5| 7	$b(\pi) = 2$
0 1 2 3 4 5 6 7	$b(\pi) = 0$

This implies:

reversal distance ≥ #breakpoints / 2

because each reversal can eliminate 2 breakpoints in the best case.

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We have denoted $b(\pi)$, the number of breakpoints in the permutation π.
When we perform a reversal, transforming π into π', we denote $b(\pi')-b(\pi)$ with Δb.

Theorem (Kececioglu and Sankoff, 1995): $b(\pi)/2 \leq \lceil b(\pi)/2 \rceil \leq d(\pi) \leq n-1$

Proof: Lower bound : a reversal can eliminate at most two breakpoints, and $d(\pi)$ is an integer.
Upper bound : we can use at most n-1 reversals to create any desired sequence.
For instance, consider the sequence of transformations applied to positions $(i \ldots \pi^{-1}(i))$ to reconstruct prefixes. (This boils down to show that at each reversal, we associate the elimination of a breakpoint)

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Ordering by reversal: a more efficient greedy algorithm

<u>BreakPointReversalSort(π)</u>

1 **while** $b(\pi) > 0$

2 Entre tous les possibles renversements, choisir le renversement r minimizant $b(\pi \bullet r)$

3 $\pi \leftarrow \pi \bullet r(i, j)$

4 **output** π

5 **return**

Problem: can this algorithm enter in some infinite loop? by eliminating some breakpoints we can, in principle, find no more breakpoints decreasing the total number.

0 1 2 | 8 9 | 5 6 7 | 3 4

0 1 2 3 | 7 6 5 | 9 8 | 4

With the strategy of augmenting the prefix, we cannot reduce the number of breakpoints

We find the same number of breakpoints